

growtoo feature specification

Product behaviour for a future team · 4 September 2026 · internal, not a marketing page

This is **what the product does** on <https://growto.live>. Stack, Firestore, Functions, and Solana live in the rebuild packet ([docs/rebuild/](#)). Root README.md is stale. Do not treat pitch-deck numbers as product facts.

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1. What growtoo is

growtoo is a **free cannabis / CBD grow journal**. The journal is the product. Reminders, a 3-day weather widget, in-app camera, an AI coach, photos, growth-stage tracking, public Stories, and an optional Solana **Devnet** proof layer (Tokenise + Market) sit on top of that journal.

It is a **documentation tool**, not legal advice, not a diagnosis service, and not a live-money marketplace. Users must be **18+** and legally allowed to use grow-related information services where they live. Growers must follow the law where they grow.

Two desks after signup:

Desk	Job	Wallet
Grower	Journal plants, log care, coach, Stories. Tokenise / Market only after chain opt-in.	Optional. Needed only to mint or list on Devnet.
Adopter	Market / garden desk. Follow listed plants, invest test \$GROWTOO, daily faucet. Journal plants are empty by default.	Needed to invest / claim faucet.

profileType (grower | adopter) is **immutable** after create on the client. Do not let users flip desks without an explicit product decision — faucet, Invest, and Tokenise are a trust boundary.

2. Status legend

Use this in every feature below.

Status	Meaning
Live	Ships to signed-in users on growto.live today.
Gated	Live, but behind a flag, allowlist, verified email, or chain opt-in.
Devnet	Live on Solana's public test network. Tokens have no monetary value .
Mocked	UI / data model exist; fulfillment is not a real pipeline.
Planned	Roadmap only. Do not announce dates.

Not live (do not claim): mainnet; \$GROWTOO as money; trustless proof that a physical plant exists; physical harvest redemption / fulfillment.

On-chain records on Devnet are **process evidence from the journal trail**, not proof a plant exists.

3. Surfaces a user hits

Surface	URL	Who
Marketing landing	/ (/hr/, /de/)	Public
Sign-in / sign-up	/dnevnik/	Public → Auth
Signed-in app	/app/	Grower / Adopter / Admin
Public Stories feed	https://journal.growto.live (also /journal/ on apex)	Public, no account
Public grower profile	journal.growto.live/g/{slug}	Public
Public story	journal.growto.live/p/{slug}	Public
Terms / Privacy / Risks / RWA docs	/terms/, /privacy/, /risks/, /rwa-docs/	Public
Pitch chooser	/pitch/	Public (grower, adopter, investor, skeptic, labs)

Languages: **English** (source of truth), **Croatian**, **German**. App switches in place (?lang= + localStorage['growtoo:lang']). Missing copy falls back to English, never a key name.

Internal data keys stay Croatian. Do not branch on displayed labels.

- Stages: klijanje → sadnica → vegetativna → cvjetanje → susenje
 - Care types: zalijevanje, gnojidba, plus općenito, okolis, presađivanje, stresori, ostalo, faza, podfaza
-

4. Information architecture (app)

Primary bottom / sidebar nav is thumb-sized. Older docs that mention a Today tab or a separate Plants tab are stale.

Grower bar

Journal · Log · Coach · Tokenise

- **Journal** — plants + calendar / list of entries. Dashboard and Today views still exist in the DOM; navigation redirects both here.
- **Log** — FAB / sheet, not a page. Fast watering / feeding; “Full journal entry” opens the entry modal.
- **Coach** — overlay panel (grower + adopter).
- **Tokenise** — 4th tab. First tap opens Tokenise. Market is the other segment on the same screen.

More / Profile (header sheet): Appearance, Language, Stories (mobile), Measurements, Unlock chain, Privacy, Terms, Log out. Admin tile only for two allowlisted emails.

Adopter bar

Same four slots, but the 4th tab is labelled **Market**. Tokenise / Garden is the other segment. Journal plants are empty by default; Log is grower-only.

Deep links: ?view=dashboard and ?view=danas resolve to Journal.

5. Accounts, onboarding, profile

Status: Live

Sign-in

- Firebase Auth: **email/password** and **Google**.
- Password: at least 8 characters (signup hint).
- Sign-in opens Grower or Adopter workspace from stored `profileType`. No second role picker.

Sign-up

1. Age gate **18+** + Terms + Privacy (required).
2. Choose desk: Grower or Adopter. This is permanent on the client.
3. Grower extras:
 - Grow setup: indoor / outdoor / mixed (seeds plant defaults and coaching).

- Home city (optional) — weather widget only.
- Grow style note (optional).
- Profile photo / logo (optional).

4. Adopter extras:

- Intent: support growers / collect a garden / learn the flow.
- Required checkbox: Devnet assets have no monetary value.

Profile (More sheet)

- Display name, profile photo, email verification status.
- Wallet link status (not linked / linked pubkey). Clients may **unlink**. Only Cloud Function linkWallet may **set** solanaPubkey after ed25519 verify. Never let the client write the pubkey.
- Rank · XP chip (growers).
- Detail level: Simple (hide mint addresses) vs Advanced (explorer links, PDAs).
- Appearance: Light (product default) / Dark / Auto. Stored as growtoo:appearance.
- Language: EN / HR / DE.
- Replay product tour.
- Resend verification / "I already verified".

Email verification gate

Live Gemini (Coach + illness-check photos) requires `email_verified` on the ID token. Unverified accounts get the **local helper** (reminders + draft logs, no live model). After an admin verifies an email, the user must **sign out and in**.

Product tour

First-run scripts per view (Journal, Tokenise/Garden). Replay from Profile. Welcome / "While you were away" / adopter START HERE: **one intro layer per session**, never stacked with Tokenise explainers. After ≥ 30 minutes away, show the return sheet only.

6. Feature catalog

6.1 Plants (journal garden)

Status: Live · Grower

CRUD for plants in the grower's journal.

Create wizard

1. Essentials: name (required), stage.
2. Optional details: strain, start date, environment name, indoor/outdoor, photo, notes.
3. Outdoor extras: field location, planting location, daily exposure hours.

4. Advanced: count (plants in a row/cohort), subphase ladder.

Subphases (pot / field ladder)

pot_1_5dc1 → pot_51 → pot_301 → na_polju (in the field). Quick actions: next pot, to field. Stage and subphase changes write history (stageHistory, subphaseHistory, stageDates) and can emit journal notes.

Optional on create: “Also seal this on-chain” — queues a seed mint if chain is unlocked and a Devnet wallet is connected. Not required for journaling.

List

- Stage filter chips.
- Rank chip in the plants header.
- Empty garden: Coach-style next step — Add a plant or Log first watering. Do not show a mute empty calendar as the first-run state.

IDs: plant id is a stable string. Seed mints, growth mints, and market listings foreign-key this id. **Never regenerate plant ids on rewrite.**

startDate is grower-facing. Market eligibility does **not** use it (see §6.12).

6.2 Journal entries

Status: Live • Grower

A dated note on one plant (or a same-strain stack).

Field	Notes
date	Journal day YYYY-MM-DD. Display and calendar. Not used for market coverage.
type	See table below.
note	Required on the full entry form.
photo / video	Photo via camera or file. Video optional. Photos go to Storage, not inline in the state blob.
meta	Type-specific extras (transplant, stressors, stage location).
createdAt	Clock time when saved. This is what market coverage counts.
source	e.g. log sheet, toolbox, camera, coach action.

Entry types (internal keys)

Key	Label	Extra fields
opcenito	General	—
zalijevanje	Watering	Amount can come from Measurements
gnojidba	Feeding	Product / dose
okolis	Environment	Temp / RH / pH
presadjivanje	Transplanting	Soil, age, condition
stresori	Stressors	Temp, humidity, VPD, pests
faza	Stage / milestone	Optional field / planting location
ostalo	Other	—
podfaza	Sub-phase (system)	Written on subphase change

Calendar vs list

Month view is the Journal landing. List stays if the user picked it. Filter by plant. Due watering / feeding from the calendar also feeds Coach reminders — there is **no separate event store**.

Care-day rule (must keep): dumping 90 logs in one session is **one** care day. Backdating entry.date does not create extra coverage. Distinct UTC days of zalijevanje / gnojidba by createdAt count.

6.3 Log sheet (FAB)

Status: Live • Grower

Fast path from the Log tab.

- Multi-select plants. Same-strain stacks can log together.
- **Log watering / Log feeding** — one tap each.
- **Full journal entry** — opens the entry modal with those plants preselected.
- Empty garden: CTA to add a plant.

This is the primary daily habit. Coach drafts should land here in spirit (confirm-then-write), not invent a second logger.

6.4 Grow log (per-plant trail)

Status: Live • Grower

Opened from a plant card. Seed-to-harvest documentation page.

- Hero (name, stage, photo).

- Strain, environment, stage timeline.
- Photo grid + recent strip + “view all”.
- Chronological journal trail for that plant.
- Actions: new entry, edit plant, open Coach.
- Optional reward-goal line when chain activity is on.

This is the “plant passport” a grower (and later an adopter) is meant to read. Listing on Market is supposed to point at this kind of trail, not at a mint timestamp.

6.5 Measurements (toolbox)

Status: Live · Grower (soil moisture: Gated, two admin emails)

Opened from Profile → Measurements. Charts live here; each Add **also writes a matching journal entry** so there is one care story, not two ledgers.

Tool	Inputs	Chart
Watering	Date, amount (mL), plant	Amount by date
Feeding	Date, product/dose, note, plant	Feeding history
Environment	Date, °C, humidity %, pH, plant	Temp / RH / pH
Transplanting	Date, soil, age, condition, plant	List
Stressors	Date + stress fields, plant	List
Charts	Overview of the above	Combined
Soil moisture	Live sensor feed	Admin emails only

Coach snapshot includes last watering mL, last feed product, last temp/RH. If a reading is missing, Coach should ask the grower to log it rather than invent numbers.

6.6 Weather (3-day)

Status: Live · Grower

Widget on Journal. City string (not GPS). Prefills from signup homeCity. 3-day forecast for grow planning (heat, watering pace). Coach can answer “what does the forecast mean for my grow?”.

Rewrite note: WeatherAPI key is currently in client JS. Move server-side. Do not add GPS unless product asks.

6.7 Plant camera + illness check

Status: Live · Grower (live analysis: Gated — verified email)

In-app camera: preview, shutter, Flip, Gallery. After capture:

- **Log to journal** — attach photo to a plant entry.
- **Check for illness** — sends 1–2 JPEG frames through the same Coach photo path (coachChat / analyzeGrowFrames).
- **Retake.**

Rules the model and UI must keep:

- Describe what is visible first, then at most two hypotheses (pest, disease, deficiency, environment).
- Do not name a pathogen unless the photo shows matching signs.
- Never treat from the photo alone. Suggest one journal log that would confirm the top hypothesis.
- Quota: vision **25/day**. Coach chat **60/day** (UTC). Distinct copy for email_unverified, expired token, quota_exceeded.
- Permission failures: blocked vs no camera vs generic — never a silent fail. Gallery is the fallback.

Photos: Firebase Storage users/{uid}/journal/..., max ~5 MB after compression. Do not store new photos as base64 in the 1 MiB Firestore state document.

6.8 Reminders + Today card

Status: Live • Grower

Derived from journal + toolbox + weather. Not a separate calendar product.

Typical reminders:

- Watering due (pace from last logs).
- Feeding due.
- Heat check / heat + watering pace when forecast is hot and last water is stale.

Surfaces: Journal **Today** card, Coach empty state (“one next step”), Coach habit CTAs, Inbox care_due. User can dismiss a reminder id. Confirming a reminder drafts a journal action — **nothing is saved until they tap confirm.**

6.9 AI Coach

Status: Live (local helper always; Gemini Gated on verified email)

Two personalities from profileType:

| | Grower Coach | Adopter Coach |

| -- | ----- | ----- |

| Job | Next care step, drafts, photo read, tokenisation tips | Explain care progress, live stage, unlock months |

| May propose actions | Yes, max 5, user must confirm | **Never** (actions: []) |

| May mint / edit journal | Only after confirm | No |

Grower allowed actions (client confirms, then executeAction):

type	Does
create_plant	New plant (name required)
add_entry	Water / feed / general / stage / environment log
set_stage	Stage key on a known plant
import_seed	Mint/link seed for a journal plant
mint_growth	Next growth stage if journal proof exists
link_plant	Attach token to plant

No deletes. Never invent plantIds that are not in the snapshot unless creating the plant first. Ground in snapshot numbers (days in stage, last water, weather). Do not replace looking at the plant.

Caps in the UI: Today, Care, Stage, Diagnose, Weather, Tokenise.

Local helper (unverified or quota): still drafts watering/feeding from the journal. Labels replies **Live coach** vs **Local helper**.

Locales: Gemini coach copy is **en** or **hr**. German UI falls back to English for live coach. Illness-frame analysis same.

Voice: practical CBD/hemp peer, not a doctor, not a lawyer, not a yield salesperson.

6.10 Stories + public journal

Status: Live · Grower (public site is unauthenticated)

In-app **Stories** (sidebar on desktop; Profile on mobile). Grower writes posts that can publish to the public journal.

Composer: title (max 120), URL slug, category, body, optional cover photo. Draft / publish / edit / delete (in-app confirm sheet, not window.confirm).

Categories: Field note · Looking at my plants · Plant problem · Made from my plants · Tips & tricks · Visited other growers.

Public profile: publicProfileEnabled, publicSlug, publicBio, profile photo. Public path `journal.growto.live/g/{slug}`.

Public feed: anyone can read, no sign-in. CTA to start a grow / sign in. Do not leak Firebase uids, emails, or private journal entries onto this surface — only published Stories and opted-in profile fields.

Mirroring to `publicJournalPosts` / `publicGrowerProfiles` is done by Cloud Functions (Admin SDK), not by a public client write.

6.11 Inbox

Status: Live

Bell in the header. Types include: journal saved, care due, seed/growth mint success or fail, sale settled, stake received, faucet, activity bonus, harvest claim, wallet reconnect.

Empty state explains the inbox. Mark all read. Demo “load examples” exists for empty desks — do not confuse example rows with production events.

Wallet reconnect: signing out ends the **browser** wallet session; the account can stay linked to the pubkey. Copy must say that, not “wallet was stolen.”

6.12 Tokenise (optional chain)

Status: Devnet • Gated on `chainOptIn` • Grower

Journal works with **no wallet**. Tokenise / Market sit behind Unlock chain (Profile or first tap on the 4th tab). Dialog: “Unlock Tokenise & Market?” — do not dead-click.

Wallets: Phantom or Solflare, Solana **Devnet**. Link is a signed challenge verified by `linkWallet`. Challenge message is **not** translated (signature is byte-for-byte).

What Tokenise does

1. Pick a **real journal plant** (`plantId` required). Seed mint is not gated on 14-day coverage.
2. Seal seed stage → queue `seedMints` → GitHub worker mints NFT.
3. Later growth seals (`growthMints`) need **journal proof** (quests): linked plant, stage log, watering this stage; feeding optional only for germination.
4. List on Market is a **separate** gate (below).

Grower quests (checklist, not a game board): link plant, log stage, log watering, log feeding. Copy: “Journal proof complete — ready to mint.”

Rank / XP: New → Active → Dedicated → Seasoned → Master → Elite cultivator. Shown on Journal and Profile. Activity bonus can mint test \$GROWTOO for documented care (Devnet, no value).

Honesty in UI: mint toasts distinguish **queued** vs **failed**. Never claim on-chain success the queue refused. Test \$GROWTOO has no monetary value. Stage rewards copy may mention a harvest total — that is Devnet game points, not money.

6.13 Market

Status: Devnet · Gated on chain opt-in

Growers list; adopters browse / invest. Public board reads **marketPublicTape** (scrubbed: on-chain pubkeys, no Firebase uids). Full marketListings are owner/buyer, not a public collection list.

List / escrow activate (must stay in sync client + Functions)

- ≥ 14 elapsed days since first care createdAt.
- Care on $\geq 50\%$ of $\min(\text{elapsed}, 180)$ distinct UTC days.
- Watering or feeding only.
- Listings with `journalCoverage0k === false` are **hidden** from the public tape.
- Same-day mint-and-list is blocked.

Offer types

Type	Behavior
Instant sale	Plant token moves to buyer now.
Adopt stake	50% \$GROWTOO now, 50% locked until harvest-claim path.

Adopter garden: settled plants appear under My adopted plants. Watch live stage and monthly unlock status. First-run: START HERE + How to adopt; hide after first adoption.

Invest: disabled in watch-only (no signing session). Header chip shows Watch-only when the session cannot sign.

Faucet: adopter-only daily test \$GROWTOO. Growers do not get this desk.

Harvest claim: UI can file `harvestClaims` (pending). **Physical redemption is mocked** — not a fulfillment pipeline. Locked 50% release is specified, not a live warehouse / legal redemption flow.

Copy rules: no yield promises. “You can watch this plant’s trail” beats APY language. Devnet / sim traffic may appear; Coach and Market must not invent balances.

6.14 Admin

Status: Gated — two emails only

`supadmin@dneynik.live` and `admin@dneynik.live`. Firestore role is **not** enough. Ordinary growers must never see Admin nav.

Capabilities (read this as “exists today,” not as a multi-tenant SaaS spec):

- User list / login report (today / 7 days).
- Legacy plants / entries / tenants collections (admin library — **not** the grower journal blob).

- Share plants and tools read-only with another user (sharedGrants).
- Hybrid viewers: own journal editable; shared superadmin plants view-only.
- Soil-moisture tool.

Do not expand admin to “anyone with role: admin” without replacing the email allowlist on purpose.

7. Cross-cutting product rules

These are product requirements, not implementation trivia.

1. **Journal first.** Coach, reminders, Tokenise, and Market are layers. A grower who never opens a wallet must still have a complete product.
2. **Confirm before mutate.** Coach actions, destructive deletes, listings, and mints use in-app confirm sheets. No silent journal writes from the model.
3. **Two desks are a trust boundary.** Faucet and Invest are adopter-only. Tokenise mint-from-journal is grower-only. Adopter Coach cannot edit the grower’s log.
4. **Coverage is createdAt, not date.** Changing that silently rewrites who can list.
5. **Plant ids are foreign keys.** Preserve them across a rewrite.
6. **Devnet honesty** in every chain surface: no monetary value, no live harvest redemption, on-chain ≠ physical plant.
7. **Coach is advisory.** No disease diagnosis-as-fact, no legal advice, no medical claims.
8. **i18n:** EN source + HR/DE; internal keys stay Croatian; German live-coach fallback is a known gap.
9. **Light theme is the default.** Dark is opt-in.
0. **Public market board never lists raw marketListings.** Tape only.
 1. **Wallet pubkey is server-verified.** Client may unlink only.
 2. **Age 18+ and local-law compliance** stay in signup, terms, and marketing FAQ.

8. What is explicitly out of scope today

Do not build these as if they were missing bugs unless product reopens them.

- Native iOS / Android store apps (responsive web only; mockups exist under mobile/).
- Service worker / installable PWA (not in repo).
- Mainnet Solana, real \$GROWTOO value, or a published mainnet date.
- Physical harvest shipping / licensed fulfillment.
- Trustless attestation that a plant exists in the world.
- GPS weather.
- Client-mutable profileType.
- Cursor / third-party chat as the Grower Coach backend (live Coach is Gemini gemini-2.0-flash via coachChat).
- Subscription billing / SaaS paywall (journal is free).

9. Suggested test matrix for a new team

Human desk tests (wallets + verified email). Automated HTTP smoke is listed in docs/TEST_READINESS.md.

#	Path	Expect
G1	Sign up grower, skip wallet	Journal, Log, Coach local helper, weather
G2	Verify email	Live Coach + illness photo
G3	Add plant → log water/feed on Log sheet	Entries on calendar; toolbox mirror if using Measurements
G4	Camera → Log to journal	Photo on grow log, Storage URL not giant state doc
G5	Unlock chain, link Phantom Devnet, seal seed	Queue → minted; toast is queued then success/fail truthfully
G6	Try to list before 14/50%	Blocked; tape hidden
A1	Sign up adopter, faucet, browse tape	No grower journal; Invest watch-only until wallet signs
A2	Instant sale vs adopt-stake	Garden updates; harvest claim remains mocked
P1	Public journal signed-out	Stories + profiles only; no emails/uids
X1	Unverified coach photo	Local helper + verify CTA, not a generic 500
X2	Non-admin user	No Admin nav even if role is wrong

10. Related documents

Doc	Use
docs/rebuild/	Tech + data room for a rewrite kickoff
docs/TEST_READINESS.md	Desk-test nav map and smoke (nav notes from Aug 2026 still apply)
docs/FIRESTORE_RULES.md	Rules intent
docs/app-check-rollout.md	App Check is monitor mode — do not flip enforcement casually
community-agent/config/product-facts.md	What marketing is allowed to say
locales/README.md	How to add a language
Live: /rwa-docs/, /risks/, /terms/, /privacy/	Trust model and legal

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